

JOSE MIGUEL ALDERETE LOPEZ

jalderetelopez@gmail.com | +56945889842 | linkedin.com/in/jalderetelopez | github.com/Blackcito

Computer Science Engineer with a minor in Information Technologies. Profile focused on Data Science, Artificial Intelligence, Data Engineering and Infrastructure/DevOps. Experience building real-time pipelines (MariaDB, Redis), interpretable machine learning models (ANFIS, fuzzy logic) and deploying highly available Kubernetes data infrastructure with Terraform and Ansible. Ability to design comprehensive solutions, optimize databases and develop systems in complex environments.

WORK EXPERIENCE

Colchones Rosen <i>Data Analysis Intern (Professional Internship)</i>	Temuco, Chile Jul 2025 – Oct 2025
- Developed a real-time processing system integrating equipment controllers, scanners and conveyors. - Implemented an ETL pipeline with temporal correlation, operational metrics and aggregated analytics modules. - Optimized the data layer in MariaDB using CTEs, batch operations, fault-tolerant state management and Grafana dashboards.	
Catholic University of Temuco <i>Data Analyst (Initial Internship)</i>	Temuco, Chile Oct 2023 – Dec 2023
- Performed data cleaning, processing and modeling, along with executive visualizations in Tableau. - Participated in data preparation and analysis for academic projects, ensuring information accuracy.	

FEATURED PROJECTS

ANFIS – Brain Tumor Detection (Capstone Thesis) <i>Explainable tumor classification in MRI images using fuzzy logic and optimization.</i>	2023 – 2025
- Designed a 6-layer ANFIS-Sugeno engine with 7 GLCM features and 128 fuzzy rules, optimized with PSO + LSE (Python, OpenCV, PySwarm). - Built a computer vision pipeline (CLAHE, adaptive segmentation), an XAI layer for interpretability and a Tkinter interface with multilevel caching.	
Real-Time Industrial Analytics System <i>Continuous pipeline for integrating and analyzing manufacturing data.</i>	2024 – 2025
- Implemented an event-driven architecture (Redis pub/sub, MariaDB) with PLC/scanner/conveyor temporal correlation and idempotent checkpointing. - Optimized SQL queries using CTEs, window functions, batch inserts and a bit-level binary state decoder.	
Kubernetes HA – Automated High-Availability Cluster <i>HA infrastructure with Terraform, Ansible and redundant load balancers.</i>	2024
- Automated infrastructure-as-code provisioning (Terraform + Ansible) of an HA Kubernetes cluster with 3 control-plane nodes and stacked etcd. - Deployed load balancing with HAProxy + Keepalived (VRRP), Calico networking and distributed storage with Longhorn.	

EDUCATION AND CERTIFICATES

Catholic University of Temuco Computer Science Engineering (Minor in Information Technologies)	Mar 2020 – May 2025 Graduated
Certifications and Training:	
- EF SET English Certificate (76/100 – C2 Proficient) - ASR Capacitación: Data Processing in Python, Intermediate/Advanced Python, Power BI Intermediate, Power BI Basic, SAP ERP.	Advanced English

SKILLS

Python (expert). SQL (expert). MariaDB (expert). Redis (expert). NumPy (expert). Pandas (expert). Scikit-learn (expert). OpenCV (expert). ANFIS / Fuzzy Logic (expert). Kubernetes (advanced). Terraform (advanced). Ansible (advanced). Docker (advanced). ETL / Streaming (advanced). Event-driven Architecture (advanced). R (advanced). Bash (advanced). YAML / HCL (advanced). HAProxy (advanced). Keepalived (advanced). Longhorn (advanced). Git / CI-CD (advanced). Grafana (advanced). Tableau (advanced). XAI / Explainability (advanced). PSO Optimization (advanced). Power BI (intermediate). SAP ERP (intermediate).